Seismic Isolation Bearing Design

003 Shear Stiffness

Example 04 - rivtbook

proj. 0004

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0.3i | Rubber Elastic Properties

Table 1: IRHD hardness scale (MN/m2)

Hardness **IRHD +/-2**	**Young's modulus** **E, MN/m2**	**Shear modulus** **G, MN/m2**	**k**	**Bulk modulus** **B(inf), MN/m2**
30	0.92	0.30	0.93	1000
35	1.18	0.37	0.89	1000
40	1.50	0.45	0.85	1000
45	1.80	0.54	0.80	1000
50	2.20	0.64	0.73	1030
55	3.25	0.81	0.64	1090
60	4.45	1.06	0.57	1150
65	5.85	1.37	0.54	1210
70	7.35	1.73	0.53	1270
75	9.40	2.22	0.52	1330

Table 2: Shore A hardness scale (lbf/in2)

Shore A **(approx.)**	**Young's modulus** **lbf/in2**	**Shear modulus** **lbf/in2**	**k**	**Bulk modulus** **lbf/in2**
35	168	53	0.89	142 000
45	256	76	0.80	142 000
55	460	115	0.64	154 000
65	830	195	0.54	171 000
75	1340	317	0.52	189 000

Note

1. The majority of springs are in the hardness range 40-60 IRHD.

2. Theoretically, with a Poisson's ratio of 1/2, B should equal 3G. This is true for soft gum rubbers. For harder rubbers containing a fair proportion of non-rubber constituents, thixotropic and other effects increase 3G to about 4G.

0.3 - 2v | Shear Stiffness Example

Table 3: Bearing Specifications

variable	value	[value]	description
G_1	58.00 p_si	0.40 MPA	shear modulus
K_1	200000.00 p_si	1378.95 MPA	bulk modulus
rnl	51	51	number of rubber layers
rdia	39.50 inch	100.33 cm	bearing diameter
rthk	0.40 inch	1.02 cm	layer thickness
sdia	38.00 inch	96.52 cm	shim diameter
sthk	0.09 inch	0.23 cm	11 guage shim thickness
cpi	3.1418	3.1418	contant pi

Bearing Stiffness for Circular Bearing

Eq. 1: rubber height

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rht = rnl·rthk

rht = 20.40 inch     [rht] = 51.82 cm   | rubber height
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rthk	rnl
0.40 inch	51
layer thickness -	number of rubber layers

Eq. 2: bearing height

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bht = rht + sthk·(rnl - 1)
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bht = 24.95 inch   [bht] = 63.37 cm | bearing height

sthk	rht	rnl
0.09 inch	20.40 inch	51
11 guage shim thickness	rubber height -	number of rubber layers

Eq. 3: shear stiffness

$$Ks_1 = \frac{G_1 \cdot cpi \cdot rdia^2}{4 \cdot rht}$$

Ks_1 = 3.48 k_in   [Ks_1] = 6.10 kN_cm | shear stiffness

cpi	rht	G_1	rdia
3.1418	20.40 inch	58.00 p_si	39.50 inch
contant pi	rubber height	shear modulus	bearing diameter

Shape Factor for Circular Bearing

Eq. 4: shape factor 1

$$sh_{1\ 1} = \frac{sdia}{4 \cdot rthk}$$

$sh_{1\ 1}$ = 24   [$sh_{1\ 1}$] = 24 | shape factor 1

rthk	sdia
0 inch	38 inch
layer thickness	shim diameter

Eq. 5: shape factor 2

$$sh_{2\ 1} = \frac{bht}{rdia}$$

$sh_{2\ 1}$ = 1 [$sh_{2\ 1}$] = 1 | shape factor 2

rdia	bht
40 inch	25 inch
bearing diameter	bearing height

0.3 - 3i | shear

Natural Frequency

The natural frequency (n_f) in Hz of a mounted body on a spring is

n_f = 5 / sqrt(x)

where x is the effective deflexion of the spring in centimetres. The effective deflexion is affected (and usually decreases) by the amplitude effect and by non-linear springs (Fig. 16).

[Figure: fig16_static_load_deformation_curve.png] Fig. 16.–OAB is a 'static' load deformation curve. Its exact position and shape will depend on the amount of structure breakdown. The static stiffness at A is the slope of the tangent AC. When a dynamic, small amplitude deformation ED is superimposed on the static deformation at A its stiffness is the slope of AF. Thus the effective dynamic deformation FG is smaller than CG or OG, and the natural frequency will be higher than that predicted from static behaviour. C and F are generally closer to O than illustrated, especially with gum and lightly-filled natural rubber vulcanizates.

[Figure: fig17_transmissibility_graph.png] Fig. 17.–The transmissibility of a natural rubber containing 40 parts by weight of carbon black. Temperature 19 degrees C. Experimental data of Snowdon, J. C., Brit. J. Appl. Phys, 1958, 9, 461-9. The full line is the theoretical relationship for no damping.

Transmissibility

Transmissibility (T) is the ratio of the amplitude of the vibration on the 'protected' side of the spring to that of the disturbing vibration, or when the spring is on a rigid base, the ratio of the corresponding forces. T depends on the ratio of the disturbing frequency n to the natural frequency n_f of the mounted system (see above).

Under normal operating conditions when n > 3*n_f natural rubber gives attenuation comparable with materials having no damping (Fig. 17). At resonance, when n = n_f, the small amount of damping in natural rubber prevents the peak value of T becoming excessive.

Stiffness Characteristics

A mathematical theory of rubber-like elasticity has been developed for elastic deformations of up to several hundred per cent. The theory applies strictly only to ideal rubbers, which are completely reversible, incompressible and isotropic, but in practice actual rubber springs have been shown to conform quite well. The behaviour of certain kinds of rubber spring may therefore be calculated for deformations of any practicable amount.

However, when the deformation is complex, an exact solution may be unattainable. In some of these cases approximate relationships can be derived from the classical theory of elasticity. This theory, which is the basis of standard engineering practice, only applies to materials in which the strains are very small (a few per cent), but the errors introduced by extrapolation to strains of the order of 10-20% are not excessive. In the stiffness equations which follow, terms beyond the first have only been given where their contribution is likely to exceed the variability of the elastic constants. Calculated stiffnesses should be within +-15% of the actual stiffness in most cases; small adjustments can be obtained by slight changes of rubber hardness.

In general, stiffness is particular to a given direction, the stiffnesses in other directions may sometimes be an order of magnitude different as, for example, in bridge bearings. By the suitable selection and location of two or more units the stiffness of a composite spring in three different directions can be varied independently. The design of such springs, of which the inclined shear mounting is a typical example, requires only a knowledge of mechanics and the principal stiffnesses of the units.

SHEAR MOUNTING

[Figure: fig18_shear_mounting.png] Fig. 18. Shear mounting.